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Optimal designs for the Kalman filter applied to electrical power distribution grids

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Abstract We consider dynamic electrical power distribution grids where unknown voltage states are strongly time dependent. These unknown states are estimated via an extended Kalman filter based on power measurements at the nodes of the grid. The design problem is here to determine nodes of the grid where the measurements should be taken with high precision and where lower precision is possible. Since the covariance matrices of state estimation and state prediction are time dependent and are additionally of complicated form, we propose a simplified design criteria based on a simplified matrix. Although this matrix is still time dependent, it is much easier to treat and methods developed for other networks can be applied. We show this for a star network using the trace and the determinant. A simulation study demonstrates that optimal designs for these simplified criteria are also very good for minimizing the trace and the determinant of the covariance matrices of state estimation and state prediction. Besides a star network, this simulation treats also a network with series connection.

Keywords Electrical Power Distribution Grids · Extended Kalman Filter · A-Optimal Design · D-Optimal Design · Star Network · Series Connection

1 Introduction

Classical approaches for estimation of unknown voltage states in an electrical power distribution grids assume a static behavior over short time periods and the unknown states are estimated by the weighted least square estimator, see,

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e.g., Schweppe and Wildes (1970, pp. 120-125). Nowadays, loads as electrical cars or heating pumps and infeeds as photovoltaic (PV) systems demand the use of a dynamic approach where the voltage states are strongly time dependent. Recently, there are several approaches to estimate the unknown time dependent voltage states via the Kalman filter and related methods, see, e.g., Ahmad et al (2017); Cetenović and Ranković (2018); Bai et al (2021); Fotopoulou et al (2022); Wu et al (2023); Bhatti et al (2024).

However, these approaches do not consider the uncertainty in state estimation and state prediction given by the Kalman filter. Only Zanni et al (2016) considers the problem of estimating the covariance matrix of state prediction. However, they assume a linear state space model and use a special optimization problem to find the estimated covariance matrix although the Kalman filter itself provides a natural covariance matrix of state prediction in each update step. They also do not consider the problem how to minimize this covariance matrix via the precision of the measurements at the nodes of the electrical power distribution grid.

The precision of measurements of the grid can be very different. Only at few nodes, sensors providing very precise measurements are possible. To obtain identifiability of the voltage estimates, additional measurements, so called pseudo measurements, are necessary, see, e.g., Schlösser et al (2014). They are in particular provided by weather data and historical data and are much less precise.

The problem of sensor allocation in distributed systems has often been addressed in research during the past 50 years, see for example the surveys in Kubrusly and Malebranche (1985), Uciński (2022), Alexanderian (2021), Duan et al (2022) or the book of Uciński (2004). In particular, the problem of allocating F sensor or generator positions out of $E > F$ possible nodes in electrical power grids was especially treated, for example, by Li et al (2011), Xygkis et al (2018), and Cao et al (2022) by greedy algorithms since the number of positions is high. Several algorithms were also proposed for optimal designs for networks appearing in other applications, see, e.g., Koutra et al (2021), Zhang (2025), Bui et al (2025). Some theoretical results for simple networks can be found in Müller and Schorning (2023).

However, they address only the static behavior of the grid and do not consider any time dependence of measurements. Du et al (2020) provided a recursive online optimization of generators in electrical power grids, but time-dependence is not modeled. Only few approaches deal with time-dependent systems. For example, Singhal and Michailidis (2008) considered the sampling problem for routers to estimate network flow volumes and track them over time with a linear Kalman filter. Duan et al (2022) developed an approach for sensor scheduling in cyber-physical systems given linear state-space equations by reformulating the problem as a Markov decision process with stationary states. An adaptive sensor placement in monitoring a network with a linear Kalman fiter was treated in Jiang and Gómez (2025). Other approaches as those in Alexanderian (2021), Uciński (2022), Uciński and Patan (2024), Uciński (2025) considered the sensor location problem in spatio-temporal systems modeled by partial differential equations.

Here we consider the sensor location problem in electrical power distribution grids where the measurements depend nonlinear on the voltage states and the voltage states are estimated by the extended Kalman filter (EKF). According to Fotopoulou et al (2022), the EKF is classified as most widely used variation of the Kalman filter for linearization of nonlinear systems.

Section 2 provides the model and the extended Kalman filter. The design problem is presented in Section 3. The design problem is based on the covariance matrices of state prediction and state estimation in the Kalman filter which are time dependent. Since these covariances are of complicated form, we propose here a simplified matrix and show, if a design minimizes the simplified matrix in the Löwner ordering, then it minimizes all covariances appearing in the Kalman filter. Since such designs are difficult to obtain, we derive in this section also the D-optimal design for the simplified criterion.

Section 4 provides the A-optimal design for the simplified criterion in a star network, which needs more work and a further simplification. Since the simplified matrix has no interpretation as a covariance matrix in the Kalman filter, Section 5 derive A- and D-optimal designs for a simplified version of the covariance matrix of state estimation under strong assumptions. Although these assumptions are usually not satisfied, these results demonstrate that optimal designs may depend on a precision bound. Section 6 provides an example and Section 7 a simulation study for this example. The simulation study indicates that optimal designs for the simplified matrix derived in Section 3 leads to good designs for the covariance matrices of state estimation and prediction with the Kalman filter. Section 8 add a simulation study for a network with series connection where no theoretical results are possible up to now. Finally, in Section 9, a discussion of the results is given.

2 The model and the extended Kalman filter

Let $(V_t)_{t\in\{0,1,\ldots,T\}}$ be the $2N$ dimensional not observable state process of real and imaginary voltage values e^t and f^t at N nodes and $(Z_t)_{t\in\{1,2,\ldots,T\}}$ the $2N$ dimensional observed process of active and reactive power measurements P_{obs}^t and Q_{obs}^t at N nodes. A general model is then

$$V_t = \Theta_{v,t} V_{t-1} + \Theta_{x,t} x_t + E_t^v, \qquad Z_t = h_t(V_t) + E_t^z,$$

where

- $\Theta_{v,t} \in \mathbb{R}^{2N\times 2N}$ is the effect of the autoregression at time t,
- $x_t \in \mathbb{R}^D$ is the vector of D known explanatory variables as load profiles of household demand, heating pumps and e-mobility, temperature (forecast), sun shine intensity (forecast), wind intensity (forecast),
- $\Theta_{x,t} \in \mathbb{R}^{2N\times D}$ contains the effects of the D explanatory variables at time t, where the effects can be different at the N nodes,

- E_t^v is the $2N$ dimensional random vector of errors of the state vector, which is independent of $V_{1:t-1}$ and $Z_{1:t-1}$,
- E_t^z is the $2N$ dimensional random vector of observation errors, which is independent of $V_{1:t}$ and $Z_{1:t-1}$,
- $h_t : \mathbb{R}^{2N} \to \mathbb{R}^{2N}$ denotes a given link function between the unknown voltage states and the power measurements.

For simplicity, it is assumed that the link function h is not time dependent. Usually it is given by the power flow equations (Papailiou, 2021, p. 291-292) so that

$$h(v_t) : \mathbb{R}^{2N} \ni v_t = (e_1^t, f_1^t, e_2^t, f_2^t, \cdots, e_N^t, f_N^t)^\top$$
$$\longrightarrow \quad (Re(\underline{S_1^t}), Im(\underline{S_1^t}), Re(\underline{S_2^t}), Im(\underline{S_2^t}), \cdots, Re(\underline{S_N^t}), Im(\underline{S_N^t}))^\top \in \mathbb{R}^{2N}$$

with apparent power

$$\underline{S_n^t} = P_n^t + jQ_n^t = (e_n^t + j \cdot f_n^t) \cdot \sum_{k=1}^{N+1} \underline{a_{nk}}^\star (e_k^t + j \cdot f_k^t)^\star,$$

where P_n^t and Q_n^t are the active power and the reactive power without measurement error at node $n = 1, 2, \cdots, N$ and time t, $\underline{a_{nk}} = g_{nk} + jb_{nk}$ the (n,k)'th component of a known admittance matrix $\underline{A} \in \mathbb{C}^{(N+1)\times(N+1)}$ representing the grid structure, j imaginary unit, $\underline{a}^\star$ complex conjugate of $\underline{a} \in \mathbb{C}$. The node $N+1$ is the slack node with $e_{N+1}^t = 1$, $f_{N+1}^t = 0$.

The unobservable state process V_t can be estimated and predicted with the Kalman filter, as given, e.g., in (Petris et al, 2009, pp. 53). However, the original Kalman filter is defined for linear link functions h so that it must be extended with a Taylor approximation, see e.g. Fotopoulou et al (2022).

Assume $E_t^v \sim \mathcal{N}_{2N}(0_{2N}, \Sigma_t^v)$, $E_t^z \sim \mathcal{N}_{2N}(0_{2N}, \Sigma_t^z)$, and $V_0 \sim \mathcal{N}_{2N}(g_0, G_0)$. Let be $z_{1:t} := (z_1, z_2, \ldots, z_t)^\top \in \mathbb{R}^{t\times 2N}$ the realization of $(Z_1, Z_2, \ldots, Z_t)^\top$ for $t = 1, \ldots, T$ and the conditional expectation $\mathrm{E}(V_t|z_{1:t-1}) = \mathrm{E}(V_1)$ for $t = 1$. The extended Kalman filter is then recursively defined with the following steps for $t = 1, \ldots, T$:

(a) **Prediction of V_t**
$V_t|z_{1:t-1} \sim \mathcal{N}_{2N}(a_t, R_t)$ with

$$\begin{aligned} a_t &= \mathrm{E}(V_t|z_{1:t-1}) = \mathrm{E}(\Theta_{v,t}V_{t-1} + \Theta_{x,t}x_t + E_t^v|z_{1:t-1}) \\ &= \Theta_{v,t}g_{t-1} + \Theta_{x,t}x_t, \\ R_t &= \mathrm{Cov}(V_t|z_{1:t-1}) = \mathrm{Cov}(\Theta_{v,t}V_{t-1} + \Theta_{x,t}x_t + E_t^v|z_{1:t-1}) \\ &= \mathrm{Cov}(\Theta_{v,t}V_{t-1}|z_{1:t-1}) + \mathrm{Cov}(E_t^v) = \Theta_{v,t}\mathrm{Cov}(V_{t-1}|z_{1:t-1})\Theta_{v,t}^\top + \Sigma_t^v \\ &= \Theta_{v,t}G_{t-1}\Theta_{v,t}^\top + \Sigma_t^v. \end{aligned}$$

(b) **Prediction of Z_t**
$Z_t|z_{1:t-1} \sim \mathcal{N}_{2N}(b_t, S_t)$ with

$$\begin{aligned} b_t &= \mathrm{E}(Z_t|z_{1:t-1}) = \mathrm{E}(h(V_t) + E_t^z|z_{1:t-1}) = \mathrm{E}(h(V_t)|z_{1:t-1}) \\ &\approx h(\mathrm{E}(V_t|z_{1:t-1})) = h(a_t), \\ S_t &= \mathrm{Cov}(Z_t|z_{1:t-1}) = \mathrm{Cov}(h(V_t) + E_t^z|z_{1:t-1}) \\ &\approx \mathrm{Cov}(h(a_t) + h'(a_t)(V_t - a_t) + E_t^z|z_{1:t-1}) \\ &= h'(a_t)\mathrm{Cov}(V_t|z_{1:t-1})h'(a_t)^\top + \mathrm{Cov}(E_t^z) = h'(a_t)R_t\, h'(a_t)^\top + \Sigma_t^z, \end{aligned}$$

where h' is the derivative (Jacobi matrix) of h and the mean value theorem (Taylor expansion) is applied in the second approximation. Because of these approximations, there is a loss of accuracy which is usually solved by an iteration in this step leading to the Iterated Extended Kalman Filter, see e.g. Bhatti et al (2024). However, to make the approach not too complicated, we restrict here on this simple extended Kalman filter.

(c) **Update of V_t**
$V_t|z_{1:t} \sim \mathcal{N}_{2N}(g_t, G_t)$ with

$$\begin{aligned} g_t &= \mathrm{E}(V_t|z_{1:t}) = a_t + R_t\, h'(a_t)^\top S_t^{-1}(z_t - b_t) = a_t + K_t(z_t - h_t(a_t)),\\ G_t &= \mathrm{Cov}(V_t|z_{1:t}) = R_t - R_t\, h'(a_t)^\top S_t^{-1} h_t'(a_t) R_t\\ &= [\mathbb{I}_{2N\times 2N} - K_t h'(a_t)]R_t, \end{aligned}$$

where $K_t = R_t\, h_t'(a_t)^\top S_t^{-1}$ is the Kalman gain and $\mathbb{I}_{2N\times 2N}$ the $2N \times 2N$ identity matrix.

3 The design problem and general results

The design problem is to decide at which nodes the measurements need to have the high precision given by sensors and at which nodes a low precision of the measurements are enough so that pseudo measurements given by, for example, historical data or weather data are satisfying. The measurements errors should be independent so that the covariance matrix of the measurements errors are given by $\Sigma_t^z = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_{2N}^2)$ where $\mathrm{diag}(\sigma_1^2, \ldots, \sigma_{2N}^2)$ denotes the diagonal matrix with entries $\sigma_1^2, \ldots, \sigma_{2N}^2$. It is reasonable to assume that the precision of the measurement of the active and the reactive power at a node are the same so that

$$\Sigma_t^z = \mathrm{diag}(\sigma_1^2, \sigma_1^2, \sigma_2^2, \sigma_2^2, \ldots, \sigma_N^2, \sigma_N^2) = \Sigma(d) = \mathbb{D}_d^{-1}$$

with

$$\mathbb{D}_d = \mathrm{diag}(\delta_1, \delta_1, \delta_2, \delta_2, \ldots, \delta_n, \delta_n) \quad \text{for} \quad d = (\delta_1, \delta_2, \ldots, \delta_N),$$

where $\delta_n := \frac{1}{\sigma_n^2}$ is the precision of the measurement. Further, we can assume that there is a precision bound c so that $\sum_{n=1}^N \delta_n \le c$ holds and the variances given by $\sigma_1^2, \ldots, \sigma_N^2$ cannot be arbitrary small. We neglect the problem that there might be only few types of precision values and assume the following design region

$$\Delta_c := \left\{ d = (\delta_1, \ldots, \delta_N)^\top \in [0, c]^N;\ \sum_{n=1}^N \delta_n \le c \right\}.$$

The interpretation is that the precision of the measurement at a node n should be high if δ_n is high, and that the precision of the measurement at a node n can be small if δ_n is small.

The aim is to choose $d \in \Delta$ so that the covariance matrices appearing in the extended Kalman filter should be minimized. Set $\mathbb{X}_t := h'(a_t)$ and

$$C_{0,t}(d) := \mathbb{X}_t^{-1} \Sigma(d) \mathbb{X}_t^{-\top} = (\mathbb{X}_t^\top \, \mathbb{D}_d \, \mathbb{X}_t)^{-1}.$$

Note that $\mathbb{X}_t$ is only invertible if the number of power measurements and the underlying voltage values are the same. Often more measurements are available. However, we here ignore all additional measurements.

Then the covariance matrices in the extended Kalman filter are given by

$$\begin{aligned}
C_{1,t}(d) &:= \mathrm{Cov}(V_t | z_{1:t-1}) = R_t(d) = \Theta_{v,t} \, C_{3,t-1,}(d) \, \Theta_{v,t}^\top + \Sigma_t^v, \\
C_{2,t}(d) &:= \mathrm{Cov}(Z_t | z_{1:t-1}) = \mathbb{X}_t \, R_t(d) \, \mathbb{X}_t^\top + \mathbb{D}_d^{-1} = \mathbb{X}_t \, [R_t(d) + C_{0,t}(d)] \, \mathbb{X}_t^\top, \\
C_{3,t}(d) &:= \mathrm{Cov}(V_t | z_{1:t}) = R_t(d) - R_t(d) \, [R_t(d) + C_{0,t}(d)]^{-1} \, R_t(d) \\
&= \left(\mathbb{I} - R_t(d) \, [R_t(d) + C_{0,t}(d)]^{-1} \right) R_t(d) \\
&= C_{0,t}(d) \, [R_t(d) + C_{0,t}(d)]^{-1} \, R_t(d) = \left[C_{0,t}(d)^{-1} + R_t(d)^{-1} \right]^{-1}.
\end{aligned}$$

It turns out that $C_{2,t}(d)$ and $C_{3,t}(d)$ directly depend on $C_{0,t}(d)$, and $C_{1,t}(d)$ indirectly depends via $C_{3,t-1}(d)$ on it. Hence it makes sense to use $C_{0,t}(d)$ as an additional, simplified design criteria. Ignoring the time dependence and setting $\mathbb{D}_E := \Sigma(d) = \Sigma_t^z$ and $\mathbb{X} := \mathbb{X}_t := h'(a_t)$ for some target state a_t, this leads to a similar problem as the minimization of $\mathbb{X}^{-1} \, \mathbb{D}_E \, (\mathbb{X}^\top)^{-1}$ considered in Müller and Schorning (2023). The difference to Müller and Schorning (2023) is that the entries of $\mathbb{X}$ are 2×2 matrices and time dependent.

Theorem 1 *If $C_{0,t}(d^*) \leq C_{0,t}(d)$ for $t = 1, \ldots, T$ then*

$$a)\ C_{1,t}(d^*) \leq C_{1,t}(d), \quad b)\ C_{2,t}(d^*) \leq C_{2,t}(d), \quad c)\ C_{3,t}(d^*) \leq C_{3,t}(d),$$

for $t = 1, \ldots, T$.

Proof. The proof is via induction. At first note, that $C_{3,0}(d) = G_0$ so that

$$C_{1,1}(d) := R_1(d) = \Theta_{v,1} \, C_{3,0}(d) \, \Theta_{v,1}^\top + \Sigma_1^v = \Theta_{v,1} \, G_0 \, \Theta_{v,1}^\top + \Sigma_1^v$$

depends only on the prior covariance matrix G_0 and not on d. This implies the assertions for $C_{2,1}$ and $C_{3,1}$ using the last expressions for $C_{2,1}$ and $C_{3,1}$. The induction step from t to $t+1$ follows also using the last expressions for $C_{2,t+1}$ and $C_{3,t+1}$. □

Theorem 1 means that a design d^*, which would minimize $C_{0,t}(d)$ in the Löwner ordering for $t = 1, \ldots, T$ for $d \in \Delta_c$, would minimize the covariance matrices $C_{1,t}(d)$, $C_{2,t}(d)$, $C_{3,t}(d)$ appearing in the Kalman filter as well. However, it is usually not possible to minimize a matrix valued criterion. Moreover, the matrix valued criterion depend here additionally on the time t. Hence, some functional as the determinant (D-optimality) or the trace (A-optimality) should be considered. The following theorem is easy to see.

Theorem 2 (D-optimality for $C_{0,t}(d)$) *The solutions $d^* = (\delta_1^*, \ldots, \delta_N^*)$ of $\arg\min_{d \in \Delta_c} \det(C_{0,t}(d))$ is given by $\delta_1^* = \ldots = \delta_N^* = \frac{c}{N}$ for any $t = 1, \ldots, T$.*

It is nice that the D-optimal design for $C_{0,t}(d)$ does not depend on the time t. However, since it is the uniform design, it is useless for determining at which node a higher precision is necessary and where a lower precision is sufficient. Unfortunately, it seems that similar results are not possible for the determinant of $C_{1,1}(d)$, $C_{2,t}(d)$, and $C_{3,t}(d)$. Also for the trace of $C_{0,t}(d)$, there is no explicit solution in the general case. Only for the special case of a star network, an explicit solution for the trace of $C_{0,t}(d)$ is possible.

4 A-optimal design for $C_{0,t}(d)$ in a star network

Here, we neglect the time dependence of $\mathbb{X}_t = h'(a_t)$ by assuming that the voltages vector a_t given by the voltages e_n^t and f_n^t attains a reference state x given by e_n and f_n for $n = 1, \dots, N$ and we set $C_0(d) = C_{0,t}(d)$. This holds in particular for a steady state system, but also approximately for a mean reverting state process. Then, for the special case of a star net, where each node is only connected with the slack node $N+1$, it holds

$$\underline{a_{n(N+1)}} = g_{n(N+1)} + jb_{n(N+1)} = -g_{nn} - j\, b_{nn} = -\underline{a_{nn}}$$

and $\underline{a_{nk}} = 0$ for $n \in \{1, \dots, N\} \setminus \{k\}$. This means with $e_{N+1} = 1$, $f_{N+1} = 0$

$$\begin{aligned}
P_n &= e_n\,(e_{N+1}\, g_{n(N+1)} - f_{N+1}\; b_{n(N+1)}) \\
&\quad + f_n\,(e_{N+1}\, b_{n(N+1)} + f_{N+1}\; g_{n(N+1)}) + e_n^2\, g_{nn} + f_n^2\, g_{nn} \\
&= -e_n\, g_{nn} + e_n^2\, g_{nn} - f_n\; b_{nn} + f_n^2\, g_{nn}, \\
Q_n &= -e_n\,(e_{N+1}\, b_{n(N+1)} + f_{N+1}\, g_{n(N+1)}) \\
&\quad + f_n(e_{N+1}\, g_{n(N+1)} - f_{N+1}\, b_{n(N+1)}) - e_n^2\, b_{nn} - f_n^2\, b_{nn} \\
&= e_n\, b_{nn} - e_n^2\, b_{nn} - f_n\, g_{nn} - f_n^2\, b_{nn},
\end{aligned}$$

so that

$$\begin{aligned}
&\frac{\partial(P_n, Q_n)^\top}{\partial(e_n, f_n)^\top} = \begin{pmatrix} g_{nn}\,(2\,e_n - 1) & 2\,f_n\, g_{nn} - b_{nn} \\ -b_{nn}\,(2\,e_n - 1) & -2\,f_n\, b_{nn} - g_{nn} \end{pmatrix} =: A_{nn}, \\
&\frac{\partial(P_n, Q_n)^\top}{\partial(e_m, f_m)^\top} = 0_{2\times 2} \quad \text{for} \quad n \neq m,
\end{aligned}$$

implying

$$\mathbb{X} = h'(x) = \frac{\partial h}{\partial x} = \begin{pmatrix} A_{11} & 0_{2\times 2} & \dots & 0_{2\times 2} \\ 0_{2\times 2} & A_{22} & & 0_{2\times 2} \\ \vdots & & \ddots & \\ 0_{2\times 2} & 0_{2\times 2} & & A_{NN} \end{pmatrix}.$$

This yields

$$C_0(d) = (\mathbb{X}^\top \mathbb{D}_d \mathbb{X})^{-1} = \begin{pmatrix} \sigma_1^2\, A_{11}^{-1} A_{11}^{-\top} & 0_{2\times 2} & \dots & 0_{2\times 2} \\ 0_{2\times 2} & \sigma_2^2\, A_{22}^{-1} A_{22}^{-\top} & & 0_{2\times 2} \\ \vdots & & \ddots & \\ 0_{2\times 2} & 0_{2\times 2} & & \sigma_N^2\, A_{NN}^{-1} A_{NN}^{-\top} \end{pmatrix} \quad (1)$$

so that $A_{nn}^{-1} A_{nn}^{-\top}$ is needed. The following lemma provides this quantity.

Lemma 1

a) $(\det(A_{nn}))^2 = (2\,e_n - 1)^2\,(g_{nn}^2 + b_{nn}^2)^2 = (2\,e_n - 1)^2\,|\underline{a_{nn}}|^4$,

b) $A_{nn}^{-1} A_{nn}^{-\top} = \dfrac{1}{|\underline{a_{nn}}|^2\,(2\,e_n - 1)^2} \begin{pmatrix} 4\,f_n^2 + 1 & -(2\,e_n - 1)\,2\,f_n \\ -(2\,e_n - 1)\,2\,f_n & (2\,e_n - 1)^2 \end{pmatrix}$.

Proof. a) and b) follow by simple calculations using

$$A_{nn}^{-1} = \frac{1}{\det(A_{nn})} \begin{pmatrix} -2\,f_n\,b_{nn} - g_{nn} & -2\,f_n\,g_{nn} + b_{nn} \\ b_{nn}\,(2\,e_n - 1) & g_{nn}\,(2\,e_n - 1) \end{pmatrix}. \square$$

Note that Lemma 1 b) provides

$$\operatorname{tr}\left(A_{nn}^{-1}\, A_{nn}^{-\top}\right) = \frac{1}{|\underline{a_{nn}}|^2} \frac{(2e_n - 1)^2 + 4f_n^2 + 1}{(2e_n - 1)^2}. \quad (2)$$

Theorem 3 (A-optimality for $C_0(d)$) *Set* $w_n^2 := \frac{(2e_n - 1)^2 + 4f_n^2 + 1}{(2e_n - 1)^2}$ *for the reference states* $(e_n, f_n)^\top$, $n = 1, \dots, N$. *Then*

$$d^* = (\delta_1^*, \dots, \delta_N^*) = \left(\frac{1}{\sigma_1^{2*}}, \dots, \frac{1}{\sigma_N^{2*}}\right) \in \arg\min\{\operatorname{tr}(C_0(d));\ d \in \Delta_c\},$$

for some constant c is satisfied if and only if $\sigma_n^{2*} = \frac{1}{\delta_n^*} = \frac{|\underline{a_{nn}}|}{w_n \cdot c} \sum_{k=1}^N \frac{w_k}{|\underline{a_{kk}}|}$ *for* $n = 1, \dots, N$, *i.e.* σ_n^{2*} *is proportional to the absolute value of the admittance* $|\underline{a_{nn}}|$.

Proof. The equations (1) and (2) provide

$$\operatorname{tr}((\mathbb{X}^\top\, \mathbb{D}_d\, \mathbb{X})^{-1}) = \sum_{n=1}^N \sigma_n^2\, \operatorname{tr}(A_{nn}^{-1} A_{nn}^{-\top}) = \sum_{n=1}^N \sigma_n^2 \frac{w_n^2}{|a_{nn}|^2}$$

$$= \operatorname{tr} \begin{pmatrix} \sigma_1^2\, \frac{w_1^2}{|\underline{a_{11}}|^2} & 0 & \dots & 0 \\ 0 & \sigma_2^2\, \frac{w_2^2}{|\underline{a_{22}}|^2} & & 0 \\ \vdots & & \ddots & \\ 0 & 0 & & \sigma_N^2\, \frac{w_N^2}{|\underline{a_{NN}}|^2} \end{pmatrix} = \operatorname{tr}\left((X^\top D_d X)^{-1}\right)$$

with

$$X := \operatorname{diag}\left(\frac{|a_{11}|}{w_1}, \ldots, \frac{|a_{NN}|}{w_N}\right) \in \mathbb{R}^{N\times N},\ D_d := \operatorname{diag}(\delta_1, \ldots, \delta_N) \in \mathbb{R}^{N\times N},$$

and $\delta_n = \frac{1}{\sigma_n^2}$ for $n = 1, \ldots, N$. Proposition 1 in Müller and Schorning (2023) yields that

$$(\delta_1^*, \ldots, \delta_N^*) \in \arg\min\{\operatorname{tr}\left((X^\top \mathbb{D}_d X)^{-1}\right);\ (\delta_1, \ldots, \delta_N) \in (0,1)^N, \sum_{n=1}^N \delta_n = 1\}$$

if and only if

$$\delta_n^* = \frac{\sqrt{\nu_n}}{\sum_{k=1}^N \sqrt{\nu_k}} \quad \text{with} \quad \nu_n = u_n^\top X^{-\top} X^{-1} u_n \quad \text{for} \quad n = 1, \ldots, N,$$

where u_n is the nth unit vector. Here we have

$$\nu_n = u_n^\top X^{-\top} X^{-1} u_n = u_n^\top \operatorname{diag}\left(\frac{w_1^2}{|a_{11}|^2}, \ldots, \frac{w_N^2}{|a_{NN}|^2}\right) u_n = \frac{w_n^2}{|a_{nn}|^2}$$

so that

$$\delta_n^* = \frac{\frac{w_n}{|a_{nn}|}}{\sum_{k=1}^N \frac{w_k}{|a_{kk}|}}.$$

If the restriction $\sum_{n=1}^N \delta_n = 1$ is replaced by $\sum_{n=1}^N \delta_n = c$ or even $\sum_{n=1}^N \delta_n \leq c$ then δ_n^* must be multiplied by c so that

$$\sigma_n^{2*} = \frac{1}{c} \frac{\sum_{k=1}^N \frac{w_k}{|a_{kk}|}}{\frac{w_n}{|a_{nn}|}} = \frac{|a_{nn}|}{w_n \cdot c} \sum_{k=1}^N \frac{w_k}{|a_{kk}|}. \square$$

5 Optimal designs in the star network by simplifying criterion $C_{3,t}(d)$

The D- and A-optimal designs d_c^* for the simplified criterion $C_{0,t}(d)$ derived in Sections 3 and 4 depend on the precision bound c only via the multiplication with c, i.e. $d_c^* = c \cdot d_1^*$. However, is seems that this is not completely the case for the real covariance matrices $C_{1,t}(d)$, $C_{2,t}(d)$, and $C_{3,t}(d)$. To obtain some theoretical results, which show this dependence on c, we consider here a simplified version of the covariance matrix $C_{3,t}(d)$ of state estimation.

Here we make the simplifying assumption that the covariance matrix of state prediction $R_t(d) = C_{1,t}(d)$ is neither time dependent nor dependent on the design d so that $R_t(d) =: R$. Neglecting the time dependence of $\mathbb{X}_t$ by using $\mathbb{X}$ as well, we get the simplified criterion

$$\overline{C}_3(d) := R - R[R + (\mathbb{X}^\top \mathbb{D}_d \mathbb{X})^{-1}]^{-1} R.$$

Since we consider here only a star network, we can assume that all covariance matrices are of block diagonal structure. Hence $R = \mathrm{Diag}(R_1, \ldots, R_N)$ where $\mathrm{Diag}(R_1, \ldots, R_N)$ denotes the block diagonal matrix with $R_n \in \mathbb{R}^{2\times 2}$, $n = 1, \ldots, N$, in the diagonal. We make additionally the assumption that all R_n are proportional to the identity matrix $\mathbb{I}_{2\times 2}$ and that $e_n = 1$, $f_n = 0$ for $n = 1, \ldots, N$.

Theorem 4 (A-optimality for $\overline{C}_3(d)$) *Let be $R_n = \lambda_n \mathbb{I}_{2\times 2}$, $e_n = 1$, $f_n = 0$ for $n = 1, \ldots, N$, and $c \geq \left(\max_{k=1,\ldots,N} \frac{1}{\lambda_k\, |\underline{a_{kk}}|}\right) \sum_{k=1}^N \frac{1}{|\underline{a_{kk}}|} - \sum_{k=1}^N \frac{1}{\lambda_k\, |\underline{a_{kk}}|^2}$.*
a) Then it holds

$$d_* = (\delta_1^*, \ldots, \delta_N^*) \in \arg\min\{\mathrm{tr}(\overline{C}_3(d));\ d \in \Delta_c\}$$

if and only if for $n = 1, \ldots, N$

$$\delta_n^* = \delta_n^{*c} := \frac{c + \sum_{k=1}^N \frac{1}{\lambda_k\, |\underline{a_{kk}}|^2}}{\sum_{k=1}^N \frac{1}{|\underline{a_{kk}}|}}\, \frac{1}{|\underline{a_{nn}}|} - \frac{1}{\lambda_n\, |\underline{a_{nn}}|^2}.$$

b) In particular, it holds $\delta_n^{*c} = 0$ if and only if $c = \frac{1}{\lambda_n\, |\underline{a_{nn}}|} \sum_{k=1}^N \frac{1}{|\underline{a_{kk}}|} - \sum_{k=1}^N \frac{1}{\lambda_k\, |\underline{a_{kk}}|^2}$.
c) Moreover, $\lim_{c\to\infty} \frac{\delta_k^{*c}}{\delta_n^{*c}} = \frac{|\underline{a_{nn}}|}{|\underline{a_{kk}}|}$ for all $n, k = 1, \ldots, N$.

Theorem 4 means that, if $|\underline{a_{kk}}| < |\underline{a_{nn}}|$, then a c_* exists so that $\delta_k^{*c} > \delta_n^{*c}$ for all $c \geq c_*$. Hence for c large enough, the A-optimal design for $\overline{C}_3(d)$ behaves similar as the A-optimal design for $C_0(d)$ where, according to Theorem 3, the largest weights are given to the nodes with smallest $|\underline{a_{nn}}|$.

Proof of Theorem 4. Since we assume $R = \mathrm{Diag}(R_1, \ldots, R_N)$, we get with (1)

$$\begin{aligned}
\mathrm{tr}(\overline{C}_3(d)) &:= \mathrm{tr}\left(R - R\left[R + (\mathbb{X}^\top\, \mathbb{D}_d\, \mathbb{X})^{-1}\right]^{-1} R\right) \\
&= \sum_{n=1}^N \mathrm{tr}(R_n) - \sum_{n=1}^N \mathrm{tr}\left(R_n\left[R_n + \sigma_n^2\, A_{nn}^{-1} A_{nn}^{-\top}\right]^{-1} R_n\right)
\end{aligned}$$

so that minimization of $\mathrm{tr}(\overline{C}_3(d))$ is equivalent with maximization of $\sum_{n=1}^N \mathrm{tr}\left(R_n\left[R_n + \sigma_n^2\, A_{nn}^{-1} A_{nn}^{-\top}\right]^{-1} R_n\right)$. Lemma 1 b) provides for $e_n = 1$ and $f_n = 0$

$$A_{nn}^{-1} A_{nn}^{-\top} = \frac{1}{|\underline{a_{nn}}|^2}\, \mathbb{I}_{2\times 2}.$$

Since also $R_n = \lambda_n \mathbb{I}_{2\times 2}$ and setting $\delta_n = \frac{1}{\sigma_n^2}$, we have to maximize

$$\sum_{n=1}^{N} \operatorname{tr}\left(R_n \left[R_n + \sigma_n^2\, A_{nn}^{-1} A_{nn}^{-\top}\right]^{-1} R_n\right)$$
$$= \sum_{n=1}^{N} \operatorname{tr}\left(\lambda_n \left[\lambda_n + \frac{\sigma_n^2}{|\underline{a_{nn}}|^2}\right]^{-1} \lambda_n\, \mathbb{I}_{2\times 2}\right) = 2 \sum_{n=1}^{N} \frac{\lambda_n^2\, \delta_n |\underline{a_{nn}}|^2}{\lambda_n\, \delta_n |\underline{a_{nn}}|^2 + 1}.$$

Hence we consider the Lagrange function

$$L(\delta_1, \ldots, \delta_N, \lambda) = \sum_{n=1}^{N} \frac{\lambda_n^2\, \delta_n |\underline{a_{nn}}|^2}{\lambda_n\, \delta_n |\underline{a_{nn}}|^2 + 1} + \lambda \left(\sum_{n=1}^{N} \delta_n - c\right).$$

Then

$$\frac{\partial}{\partial \delta_n} L(\delta_1, \ldots, \delta_N, \lambda) = \frac{\lambda_n^2\, |\underline{a_{nn}}|^2}{\lambda_n\, \delta_n |\underline{a_{nn}}|^2 + 1} - \frac{\lambda_n^2\, \delta_n |\underline{a_{nn}}|^2\, \lambda_n\, |\underline{a_{nn}}|^2}{(\lambda_n\, \delta_n |\underline{a_{nn}}|^2 + 1)^2} + \lambda$$
$$= \frac{\delta_n^2\, \lambda\, \lambda_n^2\, |\underline{a_{nn}}|^4 + \delta_n\, 2\, \lambda\, \lambda_n\, |\underline{a_{nn}}|^2 + \lambda_n^2\, |\underline{a_{nn}}|^2 + \lambda}{(\lambda_n\, \delta_n |\underline{a_{nn}}|^2 + 1)^2} = 0$$
$$\Leftrightarrow 0 = \delta_n^2 + \delta_n\, 2\, \frac{\lambda\, \lambda_n\, |\underline{a_{nn}}|^2}{\lambda\, \lambda_n^2\, |\underline{a_{nn}}|^4} + \frac{\lambda_n^2\, |\underline{a_{nn}}|^2 + \lambda}{\lambda\, \lambda_n^2\, |\underline{a_{nn}}|^4}$$
$$\Leftrightarrow \delta_n = -\frac{\lambda\, \lambda_n\, |\underline{a_{nn}}|^2}{\lambda\, \lambda_n^2\, |\underline{a_{nn}}|^4} \pm \sqrt{\left(\frac{\lambda\, \lambda_n\, |\underline{a_{nn}}|^2}{\lambda\, \lambda_n^2\, |\underline{a_{nn}}|^4}\right)^2 - \frac{\lambda_n^2\, |\underline{a_{nn}}|^2 + \lambda}{\lambda\, \lambda_n^2\, |\underline{a_{nn}}|^4}}$$
$$\Leftrightarrow \delta_n = -\frac{1}{\lambda_n\, |\underline{a_{nn}}|^2} \pm \sqrt{-\frac{1}{\lambda}}\, \frac{1}{|\underline{a_{nn}}|}.$$

This means that λ must be negative and to get nonnegative δ_n the only solution is

$$\delta_n = -\frac{1}{\lambda_n\, |\underline{a_{nn}}|^2} + \sqrt{-\frac{1}{\lambda}}\, \frac{1}{|\underline{a_{nn}}|}.$$

Plugging this form of δ_n in $c = \sum_{n=1}^{N} \delta_n$ yields

$$\frac{1}{\lambda} = -\left(\frac{c + \sum_{n=1}^{N} \frac{1}{\lambda_n\, |\underline{a_{nn}}|^2}}{\sum_{n=1}^{N} \frac{1}{|\underline{a_{nn}}|}}\right)^2$$

so that the claimed form of δ_n^{*c} follows.

Since $c \geq \left(\max_{k=1,\ldots,N} \frac{1}{\lambda_k\, |\underline{a_{kk}}|}\right) \sum_{k=1}^{N} \frac{1}{|\underline{a_{kk}}|} - \sum_{k=1}^{N} \frac{1}{\lambda_k\, |\underline{a_{kk}}|^2}$, we also obtain $\delta_n^{*c} \geq 0$ for all $n = 1, \ldots, N$ and besides a) also b) holds. Assertion c) follows from L'Hospital's rule which provides

$$\lim_{c\to\infty} \frac{\delta_k^{*c}}{\delta_n^{*c}} = \frac{\frac{1}{\sum_{m=1}^{N} \frac{1}{|\underline{a_{mm}}|}}\, \frac{1}{|\underline{a_{kk}}|}}{\frac{1}{\sum_{m=1}^{N} \frac{1}{|\underline{a_{mm}}|}}\, \frac{1}{|\underline{a_{nn}}|}} = \frac{|\underline{a_{nn}}|}{|\underline{a_{kk}}|}. \square$$

A similar assertion is possible for the determinant of $\overline{C}_3(d)$. At first note that each R_n can be represented as $R_n = \overline{R}_n\, \overline{R}_n^\top$, since R_n is symmetric. Then we get with (1)

$$\begin{aligned}
\overline{C}_3(d) &:= R - R[R + (\mathbb{X}^\top\, \mathbb{D}_d\, \mathbb{X})^{-1}]^{-1}\, R\\
&= \operatorname{Diag}(R_1 - R_1[R_1 + \sigma_1^2\, A_{11}^{-1} A_{11}^{-\top}]^{-1}\, R_1,\\
&\qquad \ldots, R_N - R_N[R_N + \sigma_N^2\, A_{NN}^{-1} A_{NN}^{-\top}]^{-1}\, R_N)\\
&= \operatorname{Diag}\Big(\overline{R}_1 \Big[I\!I_{2\times 2} - \Big[I\!I_{2\times 2} + \overline{R}_1^{-1} \sigma_1^2\, A_{11}^{-1} A_{11}^{-\top}\, \overline{R}_1^{-\top} \Big]^{-1} \Big]\, \overline{R}_1^\top,\\
&\qquad \ldots, \overline{R}_N \Big[I\!I_{2\times 2} - \Big[I\!I_{2\times 2} + \overline{R}_N^{-1} \sigma_N^2\, A_{NN}^{-1} A_{NN}^{-\top}\, \overline{R}_N^{-\top} \Big]^{-1} \Big]\, \overline{R}_N^\top \Big)
\end{aligned}$$

so that, with $B_n := \overline{R}_n^{-1} A_{nn}^{-1} A_{nn}^{-\top}\, \overline{R}_n^{-\top}$ for $n = 1, \ldots, N$,

$$\det(\overline{C}_3(d)) = \prod_{n=1}^N \det(\overline{R}_n)^2 \prod_{n=1}^N \det\left(I\!I_{2\times 2} - \left[I\!I_{2\times 2} + \sigma_n^2\, B_n \right]^{-1} \right).$$

Lemma 2 *If* $B_n := \overline{R}_n^{-1} A_{nn}^{-1} A_{nn}^{-\top}\, \overline{R}_n^{-\top}$ *for* $n = 1, \ldots, N$, *then*

$$d_* \in \arg\min_{d \in \Delta_c} \det(\overline{C}_3(d)) \Longleftrightarrow d_* \in \arg\min_{d \in \Delta_c} \prod_{n=1}^N \frac{1}{\delta_n^2 + \det(B_n) + \delta_n \operatorname{tr}(B_n)}.$$

Proof. Note that it holds for $A, B \in \mathbb{R}^{2\times 2}$

$$\det(A + B) = \det(A) + \det(B) + \operatorname{tr}(A)\ \operatorname{tr}(B) - \operatorname{tr}(A\, B)$$

so that

$$\det\left(I\!I_{2\times 2} + \sigma_n^2\, B_n \right) = 1 + \sigma_n^4 \det(B_n) + \sigma_n^2 \operatorname{tr}(B_n). \tag{3}$$

Further, we have

$$\operatorname{tr}\left(\left[I\!I_{2\times 2} + \sigma_n^2\, B_n \right]^{-1} \right) = \frac{1}{\det(I\!I_{2\times 2} + \sigma_n^2\, B_n)} \operatorname{tr}\left(I\!I_{2\times 2} + \sigma_n^2\, B_n \right).$$

Hence we get with (3)

$$\begin{aligned}
&\det\left(I\!I_{2\times 2} - \left[I\!I_{2\times 2} + \sigma_n^2\, B_n \right]^{-1} \right)\\
&= 1 + \det\left(- \left[I\!I_{2\times 2} + \sigma_n^2\, B_n \right]^{-1} \right) + \operatorname{tr}\left(- \left[I\!I_{2\times 2} + \sigma_n^2\, B_n \right]^{-1} \right)\\
&= 1 + (-1)^2 \det\left(I\!I_{2\times 2} + \sigma_n^2\, B_n \right)^{-1} - \frac{1}{\det(I\!I_{2\times 2} + \sigma_n^2\, B_n)} \operatorname{tr}\left(I\!I_{2\times 2} + \sigma_n^2\, B_n \right)\\
&= \frac{1 + \sigma_n^4 \det(B_n) + \sigma_n^2 \operatorname{tr}(B_n) + 1 - 2 - \sigma_n^2\ \operatorname{tr}(B_n)}{1 + \sigma_n^4 \det(B_n) + \sigma_n^2 \operatorname{tr}(B_n)}\\
&= \frac{\sigma_n^4 \det(B_n)}{1 + \sigma_n^4 \det(B_n) + \sigma_n^2 \operatorname{tr}(B_n)} = \det(B_n)\, \frac{1}{\delta_n^2 + \det(B_n) + \delta_n \operatorname{tr}(B_n)}
\end{aligned}$$

so that the assertion follows. □

Theorem 5 (D-optimality for $\overline{C}_3(d)$) *Let be $R_n = \lambda_n I\!I_{2\times 2}$, $e_n = 1$, $f_n = 0$ for $n = 1, \dots, N$, and $c \geq \left(\max_{k=1,\dots,N} \frac{N}{|\underline{a_{kk}}|^2\, \lambda_k}\right) - \sum_{k=1}^N \frac{1}{|\underline{a_{kk}}|^2\, \lambda_k}$.*
a) It holds

$$
\begin{aligned}
d_* = (\delta_1^*, \dots, \delta_N^*) &\in \arg\min_{d\in\Delta_c} \det(\overline{C}_3(d)) \\
&= \arg\min_{d\in\Delta_c} \prod_{n=1}^N \frac{1}{\delta_n^2 + \det(B_n) + \delta_n \operatorname{tr}(B_n)}
\end{aligned}
$$

if and only if for $n = 1, \dots, N$

$$
\delta_n^* = \delta_n^{*c} := \frac{1}{N}\left(c + \sum_{k=1}^N \frac{1}{|\underline{a_{kk}}|^2\, \lambda_k}\right) - \frac{1}{|\underline{a_{nn}}|^2\, \lambda_n}.
$$

b) In particular, it holds $\delta_n^{*c} = 0$ if and only if $c = \frac{N}{|\underline{a_{nn}}|^2\, \lambda_n} - \sum_{k=1}^N \frac{1}{|\underline{a_{kk}}|^2\, \lambda_k}$.
c) Moreover, $\lim_{c\to\infty} \delta_n^{*c} / \frac{c}{N} = 1$ for all $n = 1, \dots, N$.

Note that Theorem 5 c) means that the D-optimal design for $\overline{C}_3(d)$ approaches the uniform design for growing precision sum c. According to Theorem 2, the uniform design is the D-optimal design for $C_{0,t}(d)$. Hence, for large c, the both criteria provide similar designs.

Proof of Theorem 5. Since the minimization problem is equivalent with the maximization problem

$$
\arg\max_{d\in\Delta_c} \sum_{n=1}^N \ln(\delta_n^2 + \det(B_n) + \delta_n \operatorname{tr}(B_n)),
$$

we consider the Lagrange function

$$
L(\delta_1, \dots, \delta_N, \lambda) := \sum_{n=1}^N \ln(\delta_n^2 + \det(B_n) + \delta_n \operatorname{tr}(B_n)) + \lambda(\sum_{n=1}^N \delta_n - c).
$$

Then

$$
\begin{aligned}
&\frac{\partial}{\partial \delta_n} L(\delta_1, \dots, \delta_N, \lambda) = \frac{2\delta_n + \operatorname{tr}(B_n)}{\delta_n^2 + \det(B_n) + \delta_n \operatorname{tr}(B_n)} + \lambda = 0 \\
&\Leftrightarrow \delta_n^2 + \delta_n \frac{2 + \lambda \operatorname{tr}(B_n)}{\lambda} + \frac{\operatorname{tr}(B_n) + \lambda \det(B_n)}{\lambda} = 0 \\
&\Leftrightarrow \delta_n = \frac{-(2 + \lambda \operatorname{tr}(B_n)) \pm \sqrt{4 + \lambda^2(\operatorname{tr}(B_n)^2 - 4\det(B_n))}}{2\lambda}.
\end{aligned}
$$

Since $R_n = \lambda_n I\!I_{2\times 2}$, Lemma 1 provides for $e_n = 1$ and $f_n = 0$,

$$
\det(B_n) = \frac{1}{|\underline{a_{nn}}|^4} \frac{1}{\lambda_n^2}, \quad \operatorname{tr}(B_n) = \frac{1}{|\underline{a_{nn}}|^2} \frac{2}{\lambda_n},
$$

and thus $\operatorname{tr}(B_n)^2 - 4\det(B_n) = 0$. This implies

$$\delta_n = \frac{-(2 + \lambda \operatorname{tr}(B_n)) \pm 2}{2\lambda}.$$

The only solution, which can be nonnegative, is

$$\delta_n = \frac{-4 - \lambda \operatorname{tr}(B_n)}{2\lambda} = \frac{-2}{\lambda} + \frac{-1}{|\underline{a_{nn}}|^2 \lambda_n} \geq 0 \Leftrightarrow \frac{-1}{\lambda} \geq \frac{1}{2\,|\underline{a_{nn}}|^2 \lambda_n},$$

which means that λ must be negative. To determine λ set

$$c = \sum_{n=1}^{N} \delta_n = \sum_{n=1}^{N} \left(\frac{-2}{\lambda} + \frac{-1}{|\underline{a_{nn}}|^2 \lambda_n} \right) \Leftrightarrow \lambda = \frac{-2N}{c + \sum_{n=1}^{N} \frac{1}{|\underline{a_{nn}}|^2 \lambda_n}},$$

which provides the claimed form of δ_n^{*c}. Moreover, $c \geq \left(\max_{k=1,\ldots,N} \frac{N}{|\underline{a_{kk}}|^2 \lambda_k} \right)$ $- \sum_{k=1}^{N} \frac{1}{|\underline{a_{kk}}|^2 \lambda_k}$ implies $\delta_n^{*c} \geq 0$ and thus also assertion b). Assertion c) follows again with L'Hospital's rule. □

6 Example of a star network with three nodes and a slack node

Consider the special case of three nodes $1, 2, 3$ and a slack 4 node as shown in Figure 1.

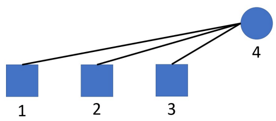

Fig. 1 Star network with three nodes $1, 2, 3$ and the slack node 4.

We have here $\underline{S_n^t} = P_n^t + jQ_n^t = (e_n^t + j \cdot f_n^t) \cdot \sum_{k=1}^{4} \underline{a_{nk}}^\star (e_k^t + j \cdot f_k^t)^\star$ for $n = 1, 2, 3$ and $\underline{S_4^t} = -(\underline{S_1^t} + \underline{S_2^t} + \underline{S_3^t} + \text{line loss})$ with voltage states $V_t = (e_1^t, f_1^t, e_2^t, f_2^t, e_3^t, f_3^t)^\top$, $e_4^t = 1, f_4^t = 0$, and the power measurements $Z_t = (P_1^t, Q_1^t, P_2^t, Q_2^t, P_3^t, Q_3^t)^\top + E_t^z$. The admittance matrix $\underline{A} \in \mathbb{C}^{4\times 4}$ was obtained from the SimBench data set of Meinecke et al (2020) as

$$\underline{A} = \begin{pmatrix} \underline{a_{11}} & 0 & 0 & \underline{a_{14}} \\ 0 & \underline{a_{22}} & 0 & \underline{a_{24}} \\ 0 & 0 & \underline{a_{33}} & \underline{a_{34}} \\ \underline{a_{41}} & \underline{a_{42}} & \underline{a_{43}} & \underline{a_{44}} \end{pmatrix} = \begin{pmatrix} \underline{a_{11}} & 0 & 0 & -\underline{a_{11}} \\ 0 & \underline{a_{22}} & 0 & -\underline{a_{22}} \\ 0 & 0 & \underline{a_{33}} & -\underline{a_{33}} \\ -\underline{a_{11}} & -\underline{a_{22}} & -\underline{a_{33}} & \underline{a_{11}} + \underline{a_{22}} + \underline{a_{33}} \end{pmatrix}$$

with $\underline{a_{11}} = 31.7 - j12.4,\ \ \underline{a_{22}} = 259.3 - j101.3,\ \ \underline{a_{33}} = 817.3 - j319.5.$

Then, according to Theorem 3, the optimal variances σ_1^{2*}, $\sigma_2^{2*}, \sigma_3^{2*}$ for the criterion $\mathrm{tr}(C_0(d))$ at the three nodes are proportional to

$$|\underline{a_{11}}| = \sqrt{31.7^2 + 12.4^2} = 34.04, \quad |\underline{a_{22}}| = \sqrt{259.3^2 + 101.3^2} = 278.38,$$
$$|\underline{a_{33}}| = \sqrt{817.3^2 + 319.5^2} = 877.53$$

so that the smallest variance and thus the highest precision should be at the node with the longest line length to the slack node, i.e., at node 1. If we set $c = 1$ and assume $(e_n, f_n) = (1, 0)$, then $w_n^2 = 2$ for $n = 1, 2, 3$ and the A-optimal design for $C_0(d)$ is

$$\delta^*_{\mathrm{tr}(C_0(d))} = \left(\frac{\frac{1}{|\underline{a_{11}}|}}{\sum_{k=1}^3 \frac{1}{|\underline{a_{kk}}|}}, \frac{\frac{1}{|\underline{a_{22}}|}}{\sum_{k=1}^3 \frac{1}{|\underline{a_{kk}}|}}, \frac{\frac{1}{|\underline{a_{33}}|}}{\sum_{k=1}^3 \frac{1}{|\underline{a_{kk}}|}} \right)$$
$$= (0.8613, 0.1053, 0.0334)^\top.$$

According to Theorem 2, the criterion $\det(C_{0,t}(d))$ leads to equal variances, i.e. $\sigma_1^{2*} = \sigma_2^{2*} = \sigma_3^{2*}$. Taking $c = 1$, the D-optimal design for $C_{0,t}(d)$ is then

$$d_1 := \delta^*_{\det(C_{0,t}(d))} = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right)^\top.$$

7 Simulation study

The above A-optimal design for $C_0(d)$ assumes $(e_n, f_n) = (1, 0)$ for the reference state and thus $w_n^2 = 2$ for $n = 1, 2, 3$. But (e_n^t, f_n^t) is time dependent and will change during the state process so that $\mathrm{tr}(C_{0,t}(d))$ and $\det(C_{0,t}(d))$ are time dependent via $\mathbb{X}_t = h'(a_t)$. Hence the question is whether the above A-optimal design for $C_0(d)$ is indeed better than the D-optimal design for $C_{0,t}(d)$ during changing values of $\mathbb{X}_t = h'(a_t)$ based on (e_n^t, f_n^t), $n = 1, \ldots, N$, in the state process.

Another problem is that $C_{0,t}(d)$ is a simplified matrix with no interpretation as a covariance matrix. Hence, the other question is how the $\mathrm{tr}(C_0(d))$-optimal design and the $\det(C_{0,t}(d))$-optimal design behave for the other criteria $\mathrm{tr}(C_{1,t}(d))$, $\mathrm{tr}(C_{2,t}(d))$, $\mathrm{tr}(C_{3,t}(d))$, $\det(C_{2,1}(d))$, $\det(C_{2,1}(d))$, $\det(C_{3,1}(d))$, which are based on the covariance matrices appearing in the Kalman filter.

To answer these questions, a simulation study with simplified admittance values and a simple state space model was done using R (R Core Team (2022)). The used admittance values are rounded versions of the above admittance values given by $\underline{a_{11}} = 30 - j10$, $\underline{a_{22}} = 260 - j100$, $\underline{a_{33}} = 820 - j310$ so that the $\mathrm{tr}(C_0(d))$-optimal design is

$$d_0 := \delta^*_{\mathrm{tr}(C_0(d))} = (0.8699, 0.0987, 0.0314)^\top.$$

The simple state space model is

$$\begin{aligned}
&V_t = V_{t-1} + 0.9\,(v_0 - V_{t-1}) + E_t^v = 0.1\,V_{t-1} + 0.9\,v_0 + E_t^v,\\
&E_t^v \sim \mathcal{N}_6(0_6, 0.1\,\mathbb{I}_{6\times 6}),\\
&E_t^z \sim \mathcal{N}_6\left(0_6, \operatorname{diag}\left((\delta_1^*)^{-1}, (\delta_1^*)^{-1}, (\delta_2^*)^{-1}, (\delta_2^*)^{-1}, (\delta_3^*)^{-1}, (\delta_3^*)^{-1}\right)\right),
\end{aligned}$$

where $(\delta_1^*, \delta_2^*, \delta_3^*) = d_0$ or $(\delta_1^*, \delta_2^*, \delta_3^*) = d_1$. This state process is a mean reverting process where $v_0 = (0.9,\, 0.1,\, 0.9,\, 0.1,\, 0.9,\, 0.1)^\top$ is the mean to which the process is reverting. Hence, the state process varies around v_0.

The simulated state space processes starts also at this v_0 and runs for $t = 1, \ldots, 100$. This was done for the two designs and thus two covariance matrices of E_t^z with the same seed. The Kalman filter starts for both processes at $g_0 = v_0$ and $G_0 = 0.1\,\mathbb{I}_{6\times 6}$.

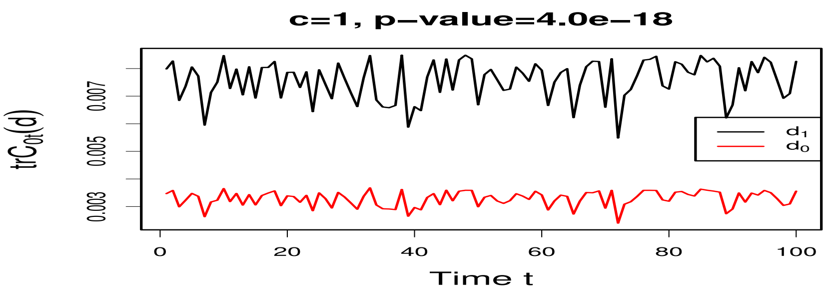

Fig. 2 Comparison of $\operatorname{tr}(C_{0,t}(d))$ (denoted here as $\operatorname{tr} C_{0t}(d))$) using the A-optimal design d_0 for $C_0(d)$ and the D-optimal design d_1 for $C_{0,t}(d)$. The shown p-value is that of the two-sided one-sample Wilcoxon test applied to the differences $\operatorname{tr}(C_{0,t}(d_1)) - \operatorname{tr}(C_{0,t}(d_0))$, $t = 1, \ldots, 100$.

Figure 2 provides the result for $\operatorname{tr}(C_{0,t}(d))$. The result for $\operatorname{tr}(C_{3,t}(d))$ is almost the same and for $\operatorname{tr}(C_{1,t}(d))$ very similar so that they are not presented here. All three cases show a time dependence. However, the A-optimal design d_0 provides a much smaller trace of $C_{0,t}(d)$, $C_{1,t}(d)$, and $C_{3,t}(d)$ than the D-optimal design d_1 at all time points. Hence the optimality of the design d_0 for the trace of the simplified matrix $C_0(d)$ transfers to the traces of the covariance matrix $C_{1,t}(d)$ of state prediction and $C_{3,t}(d)$ of state estimation.

A different result is obtained for the trace of the covariance matrix $C_{2,t}(d)$ of measurement prediction. Here the D-optimal design d_1 provides the smaller trace. This is shown in Figure 3. However, the variation over time is so large that the difference is not visible by plotting $\operatorname{tr}(C_{2,t}(d_1))$ and $\operatorname{tr}(C_{2,t}(d_0))$ separately in the upper part of Figure 3. Only a plot of the differences $\operatorname{tr}(C_{2,t}(d_1)) - \operatorname{tr}(C_{2,t}(d_0))$ in the lower part of Figure 3 shows a tendency for negative values meaning that $\operatorname{tr}(C_{2,t}(d_1))$ is smaller than $\operatorname{tr}(C_{2,t}(d_0))$. These differences are indeed significantly less than zero since the two-sided one-sample Wilcoxon test applied to the differences has a p-value of $1.2 \cdot 10^{-04}$.

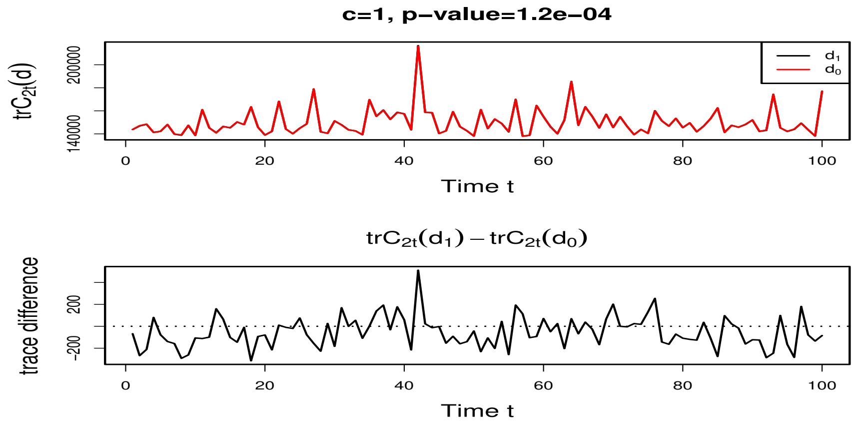

Fig. 3 Comparison of $\operatorname{tr}(C_{2,t}(d))$ (denoted here as $\operatorname{tr} C_{2t}(d)$) using the A-optimal design d_0 for $C_0(d)$ and the D-optimal design d_1 for $C_{0,t}(d)$. The shown p-value is that of the two-sided one-sample Wilcoxon test applied to the differences $\operatorname{tr}(C_{2,t}(d_1))-\operatorname{tr}(C_{2,t}(d_0))$, $t = 1,\ldots,100$.

The results of Figures 2 and 3 were obtained for a precision bound $c = 1$. Table 1 shows very similar results for precision bounds of 1 000, 100, 10, 1, 0.1, 0.01, 0.001. The only exception is $c = 0.001$, where the differences $\operatorname{tr}(C_{1,t}(d_1))-\operatorname{tr}(C_{1,t}(d_0))$ and $\operatorname{tr}(C_{3,t}(d_1))-\operatorname{tr}(C_{3,t}(d_0))$ are significantly smaller than zero.

Table 1 Medians of the differences $\operatorname{tr}(C_{i,t}(d_1)) - \operatorname{tr}(C_{i,t}(d_0))$ for $t = 1,\ldots,100$ and corresponding p-values of the two-sided one-sample Wilcoxon test in brackets for $i = 0, 1, 2, 3$ for different precision parameters c. The A-optimal design for $C_0(d)$ is d_0 and the D-optimal design for $C_{0,t}(d)$ is d_1.

Median differences $\operatorname{tr}(C_{i,t}(d_1)) - \operatorname{tr}(C_{i,t}(d_0))$

c	$\operatorname{tr}(C_{0,t}(d))$	$\operatorname{tr}(C_{1,t}(d))$	$\operatorname{tr}(C_{2,t}(d))$	$\operatorname{tr}(C_{3,t}(d))$
1e+03	4.4e-06	4.3e-08	-0.33	4.4e-06
	(4.0e-18)	(5.8e-18)	(0.72)	(4.0e-18)
1e+02	4.4e-05	4.3e-07	-1.52	4.4e-05
	(4.0e-18)	(5.8e-18)	(0.95)	(4.0e-18)
1e+01	4.4e-04	4.30e-06	-9.60	4.3e-04
	(4.0e-18)	(5.8e-18)	(0.30)	(4.0e-18)
1e+00	4.4e-03	4.14e-05	-0.77	4.1e-03
	(4.0e-18)	(5.8e-18)	(1.2e-04)	(4.0e-18)
1e-01	4.4e-02	2.6e-04	-6.9e+02	2.6e-02
	(4.0e-18)	(5.8e-18)	(1.1e-15)	(4.0e-18)
1e-02	0.45	1.3e-04	-6.5e+03	1.3e-02
	(4.0e-18)	(5.8e-18)	(4.1e-18)	(4.0e-18)
1e-03	4.52	-1.1e-03	-6.7e+04	-0.11
	(4.0e-18)	(5.8e-18)	(4.0e-18)	(4.0e-18)

Table 2 Medians of the differences $\ln\det(C_{i,t}(d_1)) - \ln\det(C_{i,t}(d_0))$ for $t = 1, \ldots, 100$ and corresponding p-values of the two-sided one-sample Wilcoxon test in brackets for $i = 0, 1, 2, 3$ for different precision parameters c. The A-optimal design for $C_0(d)$ is d_0 and the D-optimal design for $C_{0,t}(d)$ is d_1.

Median differences $\ln\det(C_{i,t}(d_1)) - \ln\det(C_{i,t}(d_0))$

c	$\ln\det(C_{0,t}(d))$	$\ln\det(C_{1,t}(d))$	$\ln\det(C_{2,t}(d))$	$\ln\det(C_{3,t}(d))$
1e+03	-5.24 (4.0e-18)	4.3e-07 (5.8e-18)	4.1e-05 (6.0e-02)	-5.24 (4.0e-18)
1e+02	-5.24 (4.0e-18)	4.3e-06 (5.8e-18)	4.5e-04 (8.8e-05)	-5.24 (4.0e-18)
1e+01	-5.24 (4.0e-18)	4.3e-05 (5.8e-18)	4.4e-03 (5.0e-14)	-5.25 (4.0e-18)
1e+00	-5.24 (4.0e-18)	4.1e-04 (5.8e-18)	4.2e-02 (4.1e-18)	-5.28 (4.0e-18)
1e-01	-5.23 (4.0e-18)	2.6e-03 (5.8e-18)	0.33 (4.0e-18)	-5.57 (4.0e-18)
1e-02	-5.24 (4.0e-18)	1.3e-03 (5.8e-18)	1.03 (4.0e-18)	-6.27 (4.0e-18)
1e-03	-5.30 (4.0e-18)	-1.1e-02 (5.8e-18)	3.6e-03 (0.45)	-5.26 (4.0e-18)

Table 2 provides the results for the determinant of the matrices $C_{0,t}(d)$, $C_{1,t}(d)$, $C_{2,t}(d)$, $C_{3,t}(d)$. We use the natural logarithm of the determinant to get not too extreme values. Again we see that $\ln\det(C_{0,t}(d))$ and $\ln\det(C_{3,t}(d))$ behave very similar and that here the D-optimal design d_1 for $C_{0,t}(d)$ is clearly the better design. However, for $\ln\det(C_{1,t}(d))$ and $\ln\det(C_{2,t}(d))$, the significantly better design is the A-optimal design d_0 for $C_0(d)$. The only exceptions are $c = 0.001$ for $\ln\det(C_{1,t}(d))$ as well as $c = 1\,000$ and $c = 0.001$ for $\ln\det(C_{2,t}(d))$ where the results concerning $c = 1\,000$ and $c = 0.001$ for $\ln\det(C_{2,t}(d))$ are not significant.

Note that always the two-sided one-sample Wilcoxon test was applied to the differences although it is not clear whether the differences can be regarded as realizations of independent random variables. However, the Ljung-Box test applied to the differences provided a minimum p-value of 0.07 and a mean p-value of 0.68 for lag 1 and a minimum p-value of 0.08 and a mean p-value of 0.72 for lag 10 considering all treated criteria. Hence the null hypothesis of no correlation within the differences is always not rejected.

Similar results were obtained for other state processes. For example, we studied a mean reverting state process given by $V_t = V_{t-1} + \Theta_v(v_0 - V_{t-1}) + E_t^v$ with v_0 as above, where Θ_v is a block diagonal matrix of 2×2 blocks. Additionally, we considered state processes which are converging to the zero vector by considering $V_t = 0.9\, V_{t-1} + E_t^v$ or $V_t = \Theta_v V_{t-1} + E_t^v$, where again Θ_v is a block diagonal matrix.

In all analyzed models, we found that $C_{0,t}(d)$ and $C_{3,t}(d)$ behave very similar with respect to the trace and the determinant. An exception appears only for very small precision parameters c. Moreover, the A-optimal design

d_0 for $C_0(d)$ is the better design for $\mathrm{tr}(C_{1,t}(d))$, and surprisingly also for $\ln\det(C_{1,t}(d))$ and $\ln\det(C_{2,t}(d))$ for many precision parameters c.

Conversely, the D-optimal design d_1 for $C_{0,t}(d)$ is the better design for $\mathrm{tr}(C_{2,t}(d))$ for many precision parameters c. An explanation for this behavior is the form of $C_{2,t}(d) := \mathbb{X}_t\, R_t(d)\, \mathbb{X}_t^\top + \mathbb{D}_d^{-1}$. If $\mathbb{D}_d^{-1}$ is large compared to $\mathbb{X}_t\, R_t(d)\, \mathbb{X}_t^\top$, which happens in particular if c is very small, then $\mathbb{D}_d^{-1}$ is dominating and $\mathrm{tr}(\mathbb{D}_d^{-1})$ is minimized by the uniform design, i.e. by the D-optimal design d_1 for $C_{0,t}(d)$.

That exceptions for $\mathrm{tr}(C_{3,t}(d))$ appear for very small precision parameters c, may be explained by Theorem 4. Although Theorems 4 and 5 were obtained under very special assumptions, Figure 4 shows in particular strong changes of the A-optimal design for the simplified criterion $\overline{C}_3(d)$. For $c = 1$ and $c = 0.0199$, node 1 gets the dominating weight, while the weight for node 1 is zero for $c = 0.0014$. Hence for small c, the D-optimal design d_1 for $C_{0,t}(d)$, which puts lower weight on node 1 than the A-optimal design d_0 for $C_0(d)$, would be the better design. Note, that $c = 0.0014$ is the lower bound in Theorem 4, while $c = 0.0199$ is the lower bound in Theorem 5.

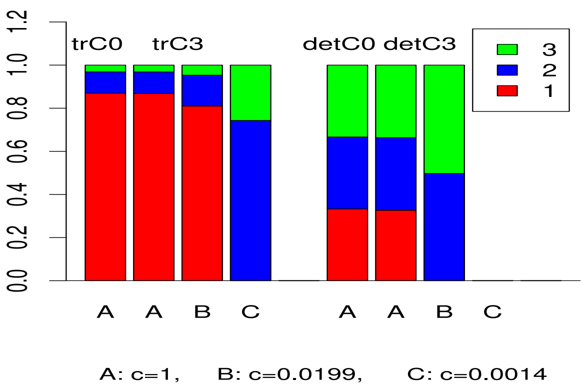

Fig. 4 Proportions of A-optimal designs at the Nodes 1, 2, 3 in the star network for $C_0(d)$ and $\overline{C}_3(d)$ (denoted here by trC0 and trC3) and the proportions of D-optimal designs for $C_{0,t}(d)$ and $\overline{C}_3(d)$ (denoted here by detC0 and detC3) for $c = 1,\ 0.0199,\ 0.0014$.

We compared also the optimal designs of Theorems 4 and 5 for small c with the A- and D-optimal designs d_0 and d_1 for $C_0(d)$ and $C_{0,t}(d)$, respectively. However, they never appeared to be better than d_0 and d_1 so that the assumptions of Theorems 4 and 5 seem to be too simplifying.

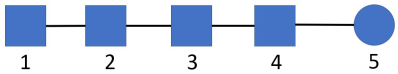

Fig. 5 Line network with four nodes $1, 2, 3, 4$ and the slack node 5.

8 Simulation study for series connection

Additionally, we consider here a grid with four nodes in a line as given in Figure 5. Then, the admittance matrix is not any more a diagonal matrix concerning the four nodes. Instead, as obtained from the SimBench data set of Meinecke et al (2020), it has the form

$$\underline{A} = \begin{pmatrix} 1.47-3.73i & -1.47+3.73i & 0.00+0.00i & 0.00+0.00i & 0.00+0.00i \\ -1.47+3.72i & 5.28-4.92i & -3.81+1.20i & 0.00+0.00i & 0.00+0.00i \\ 0.00+0.00i & -3.81+1.20i & 5.20-1.63i & -1.39+0.44i & 0.00+0.00i \\ 0.00+0.00i & 0.00+0.00i & -1.39+0.434i & 3.07-0.96i & -1.68+0.53i \\ 0.00+0.00i & 0.00+0.00i & 0.00+0.00i & -1.68+0.53i & 1.68-0.53i \end{pmatrix}.$$

Note, that here the last column and last row concern the slack node, while the first row and the first column concern the node with the largest distance to the slack node. Again, the state process is a mean reverting process with mean $v_0 = (0.9,\, 0.1,\, 0.9,\, 0.1,\, 0.9,\, 0.1,\, 0.9,\, 0.1)$ so that the model is given as

$$\begin{aligned} &V_t = V_{t-1} + 0.9\,(v_0 - V_{t-1}) + E_t^v, \quad E_t^v \sim \mathcal{N}_8(0_8, 0.1\,\mathbb{I}_{8\times 8}), \\ &Z_t = h(V_t) + E_t^z, \quad E_t^z \sim \mathcal{N}_8\left(0_8, \Sigma^z\right), \\ &\Sigma^z = \operatorname{diag}\left((\delta_1^*)^{-1}, (\delta_1^*)^{-1}, (\delta_2^*)^{-1}, (\delta_2^*)^{-1}, (\delta_3^*)^{-1}, (\delta_3^*)^{-1}, (\delta_4^*)^{-1}, (\delta_4^*)^{-1}\right), \end{aligned}$$

where $(\delta_1^*, \delta_2^*, \delta_3^*, \delta_4^*) = d_0 = (0.7,\, 0.1,\, 0.1,\, 0.1)$ or $(\delta_1^*, \delta_2^*, \delta_3^*, \delta_4^*) = d_1 = (0.1,\, 0.1,\, 0.1,\, 0.7)$. Here, d_0 provides the smallest value for $\operatorname{tr}(C_0(d))$ and d_1 the largest value for $\operatorname{tr}(C_0(d))$ within

$$\overline{\Delta}_1 = \{(0.7,\, 0.1,\, 0.1,\, 0.1), (0.1,\, 0.7,\, 0.1,\, 0.1), (0.1,\, 0.1,\, 0.7,\, 0.1), (0.1,\, 0.1,\, 0.1,\, 0.7)\}.$$

Hence, we have a sensor allocation problem here, where only one sensor, which has seven times larger precision as the other measurements, can be allocated to the nodes 1,2,3,4. It turns out that the node, which is most far away from the slack node, gets the sensor with the highest precision. Note, that $\det(C_{0,t}(d))$ attains the same value for all $d \in \overline{\Delta}_1$.

The simulated state space processes starts at $v_0 = (0.9,\, 0.1,\, 0.9,\, 0.1,\, 0.9,\, 0.1,\, 0.9,\, 0.1)^\top$ and runs for $t = 1, \ldots, 100$. As for the star network, this was done for the two designs and thus two covariance matrices of E_t^z with the same seed. The Kalman filter starts for both processes at $g_0 = v_0$ and $G_0 = 0.1\,\mathbb{I}_{8\times 8}$.

Table 3 provides the results for the trace. Again, $\mathrm{tr}(C_{0,t}(d_0))$ is significantly smaller than $\mathrm{tr}(C_{0,t}(d_1))$ for all time points. However, this transfers to $\mathrm{tr}(C_{1,t}(d))$ and $\mathrm{tr}(C_{3,t}(d))$ only for $c \leq 1$. This may be explained by the fact, that the influence of $\mathbb{D}_d^{-1}$ in $C_{2,t}(d) := \mathbb{X}_t\, R_t(d)\, \mathbb{X}_t^\top + \mathbb{D}_d^{-1}$ becomes too small for large values of c. For other mean reverting state processes with $v_0 \neq 0_8$, a similar observation was made, while no clear picture was obtained for state processes converging to the zero vector.

Table 3 Medians of the differences $\mathrm{tr}(C_{i,t}(d_1)) - \mathrm{tr}(C_{i,t}(d_0))$ for $t = 1, \ldots, 100$ and corresponding p-values of the two-sided one-sample Wilcoxon test in brackets for $i = 0, 1, 2, 3$ for the line grid. Here d_0 minimizes $\mathrm{tr}(C_0(d))$ and d_1 maximizes $\mathrm{tr}(C_0(d))$ in the sensor allocation problem.

	Median differences $\mathrm{tr}(C_{i,t}(d_1)) - \mathrm{tr}(C_{i,t}(d_0))$			
c	$\mathrm{tr}(C_{0,t}(d))$	$\mathrm{tr}(C_{1,t}(d))$	$\mathrm{tr}(C_{2,t}(d))$	$\mathrm{tr}(C_{3,t}(d))$
1e+03	9.6e-02	1.46e-04	-1.82	0.01
	(4.0e-18)	(5.8e-18)	(1.8e-17)	(4.0e-18)
1e+02	1.41	-7.5e-05	-0.34	-0.01
	(4.0e-18)	(5.8e-18)	(7.8e-18)	(4.0e-18)
1e+01	14.3	-3.3e-04	0.04	-0.03
	(5.7e-18)	(5.8e-18)	(3.3e-04)	(4.0e-18)
1e+00	1.2e+02	2.1e-05	0.07	0.002
	(4.3e-18)	(4.5e-13)	(2.3e-05)	(2.8e-13)
1e-01	1.2e+03	1.3e-04	0.03	0.01
	(4.0e-18)	(5.8e-18)	(4.0e-04)	(4.0e-18)
1e-02	1.2e+04	1.9e-05	0.01	0.002
	(4.0e-18)	(5.8e-18)	(2.1e-02)	(4.0e-18)
1e-03	1.2e+05	2.0e-06	0.002	0.0002
	(4.0e-18)	(5.8e-18)	(9.3e-02)	(4.0e-18)

9 Discussion

Since the covariance matrices appearing in the Kalman filter are of complicated form and strongly time dependent, we proposed a simplified design criterion. While D-optimal designs for this simplified design criterion are easy to derive for general networks, we could derive A-optimal designs only for the star network explicitly. Per simulation, we could show that the A-optimality of these designs transfers to most of the covariance matrices appearing in the Kalman filter in the star network. We could also show per simulation that this holds in most instances for a simple sensor allocation problem in a network with series connection. However, more simulations are necessary to study whether the simplified design criterion is useful for the quantities appearing in more complex dynamic systems. Moreover, to get A-optimal designs for the simplified design criterion in more complex dynamic systems, it should be studied whether the results of Lamberti et al (2025) for large networks with a more simple network model can be transferred to the more complicated network model considered here.

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